Chapter 1 A Tour of Computer Systems

Problem 1.1

A. We use the formula with $\alpha = \frac{3}{5}$ and $k = \frac{150}{100} = \frac{3}{2}$.

$$\begin{aligned} S &= \frac{1}{\left(1 - \frac{3}{5}\right) + \frac{3}{5} \cdot \frac{2}{3}} \\ &= \frac{1}{\frac{2}{5} + \frac{2}{5}} \\ &= \frac{5}{4} \\ &= 1.25\times \end{aligned}$$

B. We use the formula and work our way back:

$$\begin{aligned} \frac{5}{3} &= \frac{1}{\left(1 - \frac{3}{5}\right) + \frac{3}{5k}} \\ \frac{3}{5} &= \frac{2}{5} + \frac{3}{5k} \\ \frac{1}{5} &= \frac{3}{5k} \\ 1 &= \frac{3}{k} \\ k &= 3 \end{aligned}$$

So the drive through Montana needs a speedup of $3\times$ which is 300 km/hr.

Problem 1.2

Use the formula with $\alpha = \frac{4}{5}$ and $S = 2$ and solve for k.

$$\begin{aligned} 2 &= \frac{1}{\left(1 - \frac{4}{5}\right) + \frac{4}{5k}} \\ \frac{2}{5} + \frac{8}{5k} &= 1 \\ \frac{8}{5k} &= \frac{3}{5} \\ \frac{1}{k} &= \frac{3}{8} \\ k &= \frac{8}{3} \end{aligned}$$

Chapter 2 Representing and Manipulating Information

Problem 2.1

A. `0x39A7F8` to binary: `0011 1001 1010 0111 1111 1000`

B. `1100100101111011` to hexadecimal: `0xC97B`

C. `0xD5E4C` to binary: `1101 0101 1110 0100 1100`

D. `100110111001111010101` to hexadecimal: `0x26E7B5`

Problem 2.2

n	2^n (decimal)	2^n (hexadecimal)
9	512	0x200
19	524288	0x80000
14	16384	0x4000
16	65536	0x10000
17	131072	0x20000
5	32	0x20
7	128	0x80

Problem 2.3

Decimal	Binary	Hexadecimal
0	0000 0000	0x00
167	1010 0111	0xA7
62	0011 1110	0x3E
188	1011 1100	0xBC
55	0011 0111	0x37
136	1000 1000	0x88
243	1111 0011	0xF3
82	0101 0010	0x52
172	1010 1100	0xAC

Decimal	Binary	Hexadecimal
`231`	`1110 0111`	`0xE7`

Problem 2.4

A. `0x503c` + `0x8` = `0x5044`

B. `0x503c` − `0x40` = `0x4ffc`

C. `0x503c` + `64` = `0x507c`

D. `0x50ea` − `0x503c` = `0xae`

Problem 2.5

	Little endian	Big endian
A.	`21`	`87`
B.	`21 43`	`87 65`
C.	`21 43 65`	`87 65 43`

Problem 2.6

A.

`0x00359141` in binary: `0000 0000 0011 0101 1001 0001 0100 0001`

`0x4A564504` in binary: `0100 1010 0101 0110 0100 0101 0000 0100`

B.

```
00000000001101011001000101000001
  01001010010101100100010100000100
           ********************
```

There are 21 matching bits.

C.

The whole integer occurs in the float representation, except for the most-significant bit which is a 1. Similarly, some of the most-significant bits of the float representation do not occur in the int representation.

Problem 2.7

It prints `61 62 63 64 65 66` (it does not print the terminating null character because the `strlen` function does not count it).

Problem 2.8

Operation	Result
`a`	`[01101001]`
`b`	`[01010101]`
`~a`	`[10010110]`
`~b`	`[10101010]`
`a & b`	`[01000001]`
`a | b`	`[01111101]`
`a ^ b`	`[00111100]`

Problem 2.9

A. The following colors completent each other:

Black ↔ White
Blue ↔ Yellow
Green ↔ Magenta
Cyan ↔ Red

B.

Blue | Green = Cyan
Yellow & Cyan = Green
Red ^ Magenta = Blue

Problem 2.10

Step	`*x`	`*y`
Initially	`a`	`b`
Step 1	`a`	`a ^ b`
Step 2	`a ^ (a ^ b) = b`	`a ^ b`
Step 3	`b`	`b ^ (a ^ b) = a`

Problem 2.11

A. In the final iteration we have `first = k` and `last = k` (swap the middle element with itself).

B. In this case `*x` and `*y` point to the same address and the steps become:

Step	*x	*y
Initially	`a`	`a`
Step 1	`a ^ a = 0`	`a ^ a = 0`
Step 2	`0 ^ 0 = 0`	`0 ^ 0 = 0`
Step 3	`0 ^ 0 = 0`	`0 ^ 0 = 0`

C. We can fix it by changing the condition to `first < last` since the middle element does not need to be swapped anyway.

Problem 2.12

A. `x & 0xFF` leaves the least significant byte and sets everything else to zero.

B. `x ^ ~0xFF` inverts everything except the least significant byte.

C. `x | 0xFF` sets the least significant byte to ones and leaves everything else.

Problem 2.13

`x | y` is equivalent to `bis(x, y)`.

`x ^ y` is equivalent to `bis(bic(x, y), bic(y, x))`.

Problem 2.14

We have `x = 0110 0110` and `y = 0011 1001`.

Expression	Value	Expression	Value
`x & y`	`0010 0000`	`x && y`	`1`
`x | y`	`0111 1111`	`x || y`	`1`
`~x | ~y`	`1111 1111 1111 1111 1111 1111 1101 1111` (assuming 32-bit int)	`!x || !y`	`0`
`x & !y`	`0`	`x && ~y`	`1`

Problem 2.15

`!(x ^ y)` is equivalent to `x == y` because `x ^ y` will be `0` only if all the bits match.

Problem 2.16

x	x << 3	x >> 2 (logical)	x >> 2 (arithmetic)
`0xC3 = 1100 0011`	`0001 1000 = 0x18`	`0011 0000 = 0x30`	`1111 0000 = 0xF0`
`0x75 = 0111 0101`	`1010 1000 = 0xA8`	`0001 1101 = 0x1D`	`0001 1101 = 0x1D`
`0x87 = 1000 0111`	`0011 1000 = 0x38`	`0010 0001 = 0x21`	`1110 0001 = 0xE1`
`0x66 = 0110 0110`	`0011 0000 = 0x30`	`0001 1001 = 0x19`	`0001 1001 = 0x19`

Problem 2.17

Hexadecimal	Binary	$B2U_4(x)$	$B2T_4(x)$
`0xE`	`[1110]`	$2^3 + 2^2 + 2^1 = 14$	$-2^3 + 2^2 + 2^1 = -2$
`0x0`	`[0000]`	0	0
`0x5`	`[0101]`	$2^2 + 2^0 = 5$	$2^2 + 2^0 = 5$
`0x8`	`[1000]`	$2^3 = 8$	$-2^3 = -8$
`0xD`	`[1101]`	$2^3 + 2^2 + 2^0 = 13$	$-2^3 + 2^2 + 2^0 = -3$
`0xF`	`[1111]`	$2^3 + 2^2 + 2^1 + 2^0 = 15$	$-2^3 + 2^2 + 2^1 + 2^0 = -1$

Problem 2.18

A. `0x2e0 = 736`

B. `-0x58 = -88`

C. `0x28 = 40`

D. `-0x30 = -48`

E. `0x78 = 120`

F. `0x88 = 136`

G. `0x1f8 = 504`

H. `0xc0 = 192`

I. `-0x48 = -72`

Problem 2.19

x	$T2U_4(x)$
-8	8
-3	$2^3 + 2^2 + 2^0 = 13$
-2	$2^3 + 2^2 + 2^1 = 14$
-1	$2^3 + 2^2 + 2^1 + 2^0 = 15$
0	0
5	5

Problem 2.20

Equation 2.5 can be used to solve the previous problem. Since $\omega = 4$, we need to add $2^4 = 16$ to all negative numbers in Two's Complement. For example, $-8 + 16 = 8$ and $-1 + 16 = 15$. Positive numbers (and zero) stay the same.

Problem 2.21

Expression	Type	Evaluation
$-2147483647 - 1 == 2147483648U$	Unsigned	1
$-2147483647 - 1 < 2147483647$	Signed	1
$-2147483647 - 1U < 2147483647$	Unsigned	0
$-2147483647 - 1 < -2147483647$	Signed	1
$-2147483647 - 1U < -2147483647$	Unsigned	1

Problem 2.22

A. $[1011] = -2^3 + 2^1 + 2^0 = -5$
B. $[11011] = -2^4 + 2^3 + 2^1 + 2^0 = -5$
C. $[111011] = -2^5 + 2^4 + 2^3 + 2^1 + 2^0 = -5$

Problem 2.23

`w`	`fun1(w)`	`fun2(w)`
`0x00000076`	`0x00000076`	`0x00000076`
`0x87654321`	`0x00000021`	`0x00000021`
`0x000000C9`	`0x000000C9`	`0xFFFFFFC9`
`0xEDCBA987`	`0x00000087`	`0xFFFFFF87`

`fun1` keeps only the least significant byte and sets the other three to all zeroes, resulting in a value between 0 and 255. `fun2` also extracts the least significant byte, but it performs sign extension instead of zero extension, which results in a value between −128 and 127.

Problem 2.24

Hex		Unsigned		Two's complement	
Original	Truncated	Original	Truncated	Original	Truncated
0	0	0	0	0	0
2	2	2	2	2	2
9	1	9	1	−7	1
B	3	11	3	−5	3
F	7	15	7	−1	−1

We can use the equations to verify these results. For example, in hex F truncates to 7, in unsigned $B2U_4(1111) \bmod 2^3 = 7$ and in two's complement $U2T_3(B2U_4(1111) \bmod 2^3) = -1$.

Problem 2.25

Because `length` is unsigned the expression $0 - 1$ evaluates to UMax. The comparison has an unsigned integer on one side, which means the other side will also be treated as unsigned. Of course every unsigned number is $\leq$ UMax and so we try to access invalid array elements.

We can fix it by changing the condition to `i < length` or changing `length` to a signed integer.

Problem 2.26

A. The function returns wrong results in case `t` is longer than `s`.

B. The problem is that `strlen` returns a `size_t` which is unsigned. When calculating `strlen(s) - strlen(t)` where `t` is longer than `s` unsigned arithmetic is used, resulting in a number close to UMax instead of a negative number. This is obviously greater than 0 so the function incorrectly says that `s` is longer.

C. We can fix it by changing the condition to `strlen(s) > strlen(t)`.

Problem 2.27

```c
/* Determine whether arguments can be added without overflow */
int uadd_ok(unsigned x, unsigned y) {
  return x + y >= x;
}
```

Problem 2.28

x		$-_{\omega}^{u} x$	
Hex	**Decimal**	**Decimal**	**Hex**
0	0	0	0
5	5	11	B
8	8	8	8
D	13	3	3
F	15	1	1

Problem 2.29

x	y	$x + y$	$x +_5^t y$	Case
-12 [10100]	-15 [10001]	-27 [100101]	5 [00101]	1
-8 [11000]	-8 [11000]	-16 [110000]	-16 [10000]	2
-9 [10111]	8 [01000]	-1 [111111]	-1 [11111]	2
2 [00010]	5 [00101]	7 [000111]	7 [00111]	3
12 [01100]	4 [00100]	16 [010000]	-16 [10000]	4

Problem 2.30

```c
/* Determine whether arguments can be added without overflow */
int tadd_ok(int x, int y) {
  int sum = x + y;

  if (x > 0 && y > 0) {
    return sum > 0;
  }

  if (x < 0 && y < 0) {
    return sum < 0;
  }

  return 1;
}
```

Problem 2.31

sum $-\ x$ can overflow again, since it's another two's complement addition. For example, if x and y are large positive numbers whose sum overflows to a negative number, then sum $-\ x$ will cause a negative overflow "wrapping back around" to y. So this check will not detect the overflow.

Problem 2.32

The function will be incorrect for $y = \text{TMin}_\omega$. This is because the two's complement representation is not symmetric. $-y = -\text{TMin}_\omega = \text{TMin}_\omega$ causes an overflow possibly resulting in an incorrect return value.

Problem 2.33

x		$-_4^t x$	
Hex	Decimal	Decimal	Hex
0	0	0	0
5	5	−5	B
8	−8	−8	8
D	−3	3	3
F	−1	1	1

The bit patterns for two's complement and unsigned negation are the same.

Problem 2.34

Mode	x	y	$x \cdot y$	Truncated $x \cdot y$
Unsigned	$4 = [100]$	$5 = [101]$	$20 = [010100]$	$4 = [100]$

Mode	x	y	$x \cdot y$	Truncated $x \cdot y$
Two's complement	$-4 = [100]$	$-3 = [101]$	$12 = [001100]$	$-4 = [100]$
Unsigned	$2 = [010]$	$7 = [111]$	$14 = [001110]$	$6 = [110]$
Two's complement	$2 = [010]$	$-1 = [111]$	$-2 = [111110]$	$-2 = [110]$
Unsigned	$6 = [110]$	$6 = [110]$	$36 = [100100]$	$4 = [100]$
Two's complement	$-2 = [110]$	$-2 = [110]$	$4 = [000100]$	$-4 = [100]$

Problem 2.35

1. Let $t = u + p_{\omega-1}$ where u is the two's complement number represented by the ω upper bits of the 2ω-bit representation of $x \cdot y$. Since $p_{\omega-1}$ is either 0 or 1, there are two possibilities for t to equal 0.

1. If $p_{\omega-1} = 0$ then it must be that $u = 0$ (upper ω bits are all 0s).
2. If $p_{\omega-1} = 1$ then it must be that $u = -1$ (upper ω bits are all 1s).

So $t = 0$ if the upper $\omega + 1$ bits are all 0s or all 1s. These are exactly the cases where the multiplication does not overflow. All other cases do overflow.

This means we can write $x \cdot y = p + t2^{\omega}$ which overflows iff $t \neq 0$.

2. To show that p can be written in the form $p = x \cdot q + r$, where $|r| < |q|$ we consider integer division. Dividing p by nonzero x gives a quotient q and remainder r, such that $|r| < |q|$.

3. By plugging in we get $x \cdot y = x \cdot q + r + t2^{\omega}$. If $r + t2^{\omega} = 0$ then $q = y$. Since $|r| < |q| < 2^{\omega}$ this can only hold if $r = t = 0$.

Problem 2.36

```c
/* Determine whether arguments can be multiplied without overflow */
int tmult_ok(int x, int y) {
  int64_t prod = ((int64_t)x) * y;
  int64_t upper = prod >> 31;
  // if the upper 33 bits are all 1s or 0s the number fits into 32 bits
  return upper == 0 || upper == -1;
}
```

Problem 2.37

A. The new code does not improve the situation since the 64-bit number will be truncated to 32 bits when passed to `malloc`. This truncation is the same that also happens when the multiplication overflows.

B. Check the multiplication for overflow (by one of the previous methods) and if it overflows immediately abort and don't allocate any memory.

Problem 2.38

A single LEA instruction can compute the following multiples:

k	b	$(a \ll k) + b$
0	0	$(2^0 + 0)a = 1a$
0	a	$(2^0 + 1)a = 2a$
1	0	$(2^1 + 0)a = 2a$
1	a	$(2^1 + 1)a = 3a$
2	0	$(2^2 + 0)a = 4a$
2	a	$(2^2 + 1)a = 5a$
3	0	$(2^3 + 0)a = 8a$
3	a	$(2^3 + 1)a = 9a$

Problem 2.39

In this case the expression simplifies to $-(x \ll m)$. This is because shifting by $n + 1 = \omega$ to the left results in 0 so we can ignore the first term.

Problem 2.40

K	Shifts	Add/Subs	Expression
6	2	1	$(x \ll 2) + (x \ll 1)$
31	1	1	$(x \ll 5) - x$
-6	2	1	$(x \ll 1) - (x \ll 3)$
55	2	2	$(x \ll 6) - (x \ll 3) - x$

Problem 2.41

Consider two cases:

- $m > 0$: In this case form A requires $n - m + 1$ shifts and $n - m$ additions while form B requires 2 shifts and 1 subtraction. So if $n = m$, form A is favorable since it requires only 1 shift and 0 additions which is less than the constant numbers required for form B. If $n = m + 1$, form A takes 2 shifts and 1 addition, so either form is equally efficient in this case. If $n > m + 1$ form A takes $n - m + 1 > 2$ shifts and $n - m > 1$ additions so form B is favorable in this case.
- $m = 0$: In this case the last bit does not cause a shift so form A requires n shifts and n additions, while form B takes 1 shift and 1 subtraction. The same three cases apply in the same way here, so the above analysis extends to this case as well.

Problem 2.42

```
int div16(int x) {
  int bias = (x >> 31) & 15;
  return (x + bias) >> 4;
}
```

Problem 2.43

x is shifted to the left by 5 which is equivalent to multiplying by $2^5 = 32$. Then, one x is subtracted from it, so we end up with $31x$ and $M = 31$.

If y is negative, a bias of $7 = 8 - 1$ is added. Then y is shifted right arithmetically by 3 which is equivalent to dividing by $2^3 = 8$. This means that $N = 8$.

Problem 2.44

A. `false` for $x = -2^{31}$ which is `INT32_MIN`. x is obviously not greater than 0, and $x - 1$ overflows to `INT32_MAX` which is not lower than 0.

B. Always `true`. The expressions are connected by `OR` so both parts would need to evaluate to 0.

Let x_2 be x's third bit from the right. For the first part to evaluate to 0 we need $x_2 = 1$. For the second part, x_2 will become the sign so it needs to be 0 to represent a positive number. This is obviously not possible at the same time so at least one part always evaluates to 1.

C. `false` for $x = 2^{16} - 1$.

D. Always `true`. For all negative numbers the first part of the expression evaluates to 1. For 0 the second part evaluates to 1. Every positive number can be negated and still fits into a 32-bit integer so in this case also the second parts evaluates to 1.

E. `false` for $x = -2^{31}$ which is `INT32_MIN`. It's not greater than 0 and negating it causes an overflow (`-INT32_MIN = INT32_MIN`) which is still lower than 0.

F. Always `true`. Addition works the same on the bit level for both unsigned and two's complement numbers. Since the right side is unsigned, C will interpret both sides as unsigned numbers for the comparison.

G. Always `true`. `~y` is equal to $-y - 1$. Also, on the bit level `uy * ux` is the same as `x * y`. Plugging in we get $x(-y - 1) + xy = -xy - x + xy = -x$.

Problem 2.45

Fractional value	Binary representation	Decimal representation
$\frac{1}{8}$	0.001	0.125
$\frac{3}{4}$	0.11	0.75
$\frac{25}{16}$	1.1001	1.5625
$\frac{43}{16}$	10.1011	2.6875
$\frac{9}{8}$	1.001	1.125
$\frac{47}{8}$	101.111	5.875
$\frac{51}{16}$	11.0011	3.1875

Problem 2.46

A. Binary representation of $0.1 - x$ is: 0.00000000000000000000000011001100...

B. The above is the binary representation of 0.1 with the binary point shifted to the left by 20, so it is equal to $0.1 \times 2^{-20} \approx 9.54 \times 10^{-8}$.

C. After 100 hours the clock is behind by $9.54 \times 10^{-8} \cdot 100 \cdot 60 \cdot 60 \cdot 10 \approx 0.343$ seconds.

D. $2000\text{m/s} \cdot 0.343\text{s} \approx 686\text{m}$

Problem 2.47

Bits	e	E	2^E	f	M	$2^E \times M$	V	Decimal
0 00 00	0	0	1	$\frac{0}{4}$	$\frac{0}{4}$	$\frac{0}{4}$	0	0.0
0 00 01	0	0	1	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	0.25
0 00 10	0	0	1	$\frac{2}{4}$	$\frac{2}{4}$	$\frac{2}{4}$	$\frac{1}{2}$	0.5
0 00 11	0	0	1	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$	0.75
0 01 00	1	0	1	$\frac{0}{4}$	$\frac{4}{4}$	$\frac{4}{4}$	1	1.0
0 01 01	1	0	1	$\frac{1}{4}$	$\frac{5}{4}$	$\frac{5}{4}$	$\frac{5}{4}$	1.25
0 01 10	1	0	1	$\frac{2}{4}$	$\frac{6}{4}$	$\frac{6}{4}$	$\frac{3}{2}$	1.5
0 01 11	1	0	1	$\frac{3}{4}$	$\frac{7}{4}$	$\frac{7}{4}$	$\frac{7}{4}$	1.75
0 10 00	2	1	2	$\frac{0}{4}$	$\frac{4}{4}$	$\frac{8}{4}$	2	2.0
0 10 01	2	1	2	$\frac{1}{4}$	$\frac{5}{4}$	$\frac{10}{4}$	$\frac{5}{2}$	2.5
0 10 10	2	1	2	$\frac{2}{4}$	$\frac{6}{4}$	$\frac{12}{4}$	3	3.0
0 10 11	2	1	2	$\frac{3}{4}$	$\frac{7}{4}$	$\frac{14}{4}$	$\frac{7}{2}$	3.5
0 11 00	-	-	-	-	-	-	∞	-
0 11 01	-	-	-	-	-	-	NaN	-
0 11 10	-	-	-	-	-	-	NaN	-
0 11 11	-	-	-	-	-	-	NaN	-

Problem 2.48

$3'510'593$ in binary is 11 0101 1001 0001 0100 0001 which equals $1.10101100100010100000 1 \times 2^{21}$. So the biased exponent is $21 + 2^7 - 1 = 148$.

The complete single-precision floating-point representation is:

0 10010100 10101100100010100000100 or 0x4A564504

Comparing the two we see that part of it overlaps:

```
00000000001101011001000101000001
          **********************
  01001010010101100100010100000100
```

Problem 2.49

A. $2^{n+1} + 1$

B. $2^{24} + 1 = 16'777'217$

Problem 2.50

Exact		Rounded	
Binary	Decimal	Binary	Decimal
10.010_2	$2\frac{1}{4}$	10.0_2	2
10.011_2	$2\frac{3}{8}$	10.1_2	$2\frac{1}{2}$
10.110_2	$2\frac{3}{4}$	11.0_2	3
11.001_2	$3\frac{1}{8}$	11.0_2	3

Problem 2.51

A. 0.00011001100110011001101

B.

```
  0.00011001100110011001101
- 0.00011001100110011001100110011 0011 0011...
  ----------------------------------------
  0.00000000000000000000000000[1100]
```

This is equal to $\frac{1}{10} \times 2^{-22} \approx 2.38 \times 10^{-8}$.

C. After 100 hours the clock is ahead by $2.38 \times 10^{-8} \cdot 100 \cdot 60 \cdot 60 \cdot 10 \approx 0.086$ seconds.

D. $2000\text{m/s} \cdot 0.086\text{s} \approx 172\text{m}$

Problem 2.52

Format A		Format B	
Bits	Value	Bits	Value
011 0000	1	0111 000	1
101 1110	$\frac{15}{2}$	1001 111	$\frac{15}{2}$
010 1001	$\frac{25}{32}$	0110 100	$\frac{3}{4}$
110 1111	$\frac{31}{2}$	1011 000	16
000 0001	$\frac{1}{64}$	0001 000	$\frac{1}{64}$

Problem 2.53

```c
#define POS_INFIFINTY 1e400 // overflows
#define NEG_INFINITY (-POS_INFINITY)
#define NEG_ZERO (1.0/NEG_INFINITY)
```

Problem 2.54

A. Always `true` · `double` has enough range to represent all `int` values and it does not need to round. This also means no rounding when casting back.

B. `false` for $2^{24} + 1$. It is not true for all numbers with 24 or more significant bits, since `float` uses 23 bits for representing the fraction, which means these values get rounded and lose precision.

C. `false` for $2^{24} + 1$. The same reasoning as above applies here.

D. Always `true` · `double` has enough range and precision to exactly represent any `float` .

E. Always `true` . Negating the `float` only flips the sign bit and loses no information. So flipping it back results in the original value.

F. Always `true` . The integers get converted to `float` first.

G. Always `true` . If the multiplication overflows it results in $\infty > 0$.

H. `false` for $f = 10^{20}$ and $d = 1.0$. All values are promoted to `double` but even then the precision is not enough to represent $10^{20} + 1.0$. This is because $\log_2 10^{20} > 66$ and `double` has only `52` bits for representing the fraction. So when we set the least significant bit to 1 we would need about `66` bits to store the precise number. This means that adding `1.0` will have no effect on the very large number.

Problem 2.58

```c
int is_little_endian() {
  // set only least significant byte to 0x01
  uint16_t x = 1;
  // char pointer reads only first byte, so it will be:
  // 0x00 on big endian systems
  // 0x01 on little endian systems
  return *(char *)&x;
}
```

Problem 2.59

```c
int main() {
  int x = 0x89ABCDEF;
  int y = 0x76543210;
  int mask = 0xFF;
  printf("%d\n", ((x & mask) | (y & ~mask)) == 0x765432EF);
}
```

Problem 2.60

```c
unsigned replace_byte(unsigned x, int i, unsigned char b) {
  unsigned mask = ~(0xFF << (i << 3));
  return (x & mask) | (b << (i << 3));
}

int main() {
  printf("%d\n", replace_byte(0x12345678, 2, 0xAB) == 0x12AB5678);
  printf("%d\n", replace_byte(0x12345678, 0, 0xAB) == 0x123456AB);
}
```

Problem 2.61

```c
  int A = !!x;
  int B = !!~x;
  int C = !!(x & 0xFF);
  int D = !!~(x >> ((sizeof(int) - 1) << 3));
```

Problem 2.62

```c
int int_shifts_are_arithmetic() {
  return (-1 >> 1) == -1;
}
```

Problem 2.63

```c
unsigned srl(unsigned x, int k) {
  /* Perform shift arithmetically */
  unsigned xsra = (int)x >> k;

  int w = sizeof(int) << 3;
  int mask = ~(-1 << (w - k));
  return xsra & mask;
}

int sra(int x, int k) {
  /* Perform shift logically */
  int xsrl = (unsigned) x >> k;

  int w = sizeof(int) << 3;
  int mask = -1 << (w - k);
  int sign = -(x & (1 << (w - 1)));
  return xsrl | (mask & sign);
}
```

Problem 2.64

```c
int any_odd_one(unsigned x) {
  int mask = 0b10101010101010101010101010101010;
  return !!(x & mask);
}
```

Problem 2.65

```c
int odd_ones(unsigned x) {
  x ^= x >> 16;
  x ^= x >> 8;
  x ^= x >> 4;
  x ^= x >> 2;
  x ^= x >> 1;
  return x & 1;
}
```

Problem 2.66

```
int leftmost_one(unsigned x) {
  // Spread leftmost 1 to the right
  x |= x >> 1;
  x |= x >> 2;
  x |= x >> 4;
  x |= x >> 8;
  x |= x >> 16;
  return x ^ (x >> 1);
}
```

Problem 2.67

Shifting by an amount $\geq$ the width of the type is undefined for signed integers.

```
int int_size_is(unsigned w) {
  unsigned set_msb = 1U << (w - 1);
  unsigned beyond_msb = set_msb << 1;
  return set_msb && !beyond_msb;
}
```

Problem 2.68

```
int lower_one_mask(int n) {
  int w = sizeof(int) << 3;
  return ~0U >> (w - n);
}
```

Problem 2.69

```
unsigned rotate_left(unsigned x, int n) {
  unsigned w = sizeof(unsigned) << 3;
  unsigned rotated_bits = x >> (w - n - 1) >> 1;
  return (x << n) | rotated_bits;
}
```

Problem 2.70

```
int fits_bits(int x, int n) {
  int w = sizeof(int) << 3;
  int shifted = x << (w - n) >> (w - n);
  return shifted == x;
}
```

Problem 2.71

A. The upper three bytes are always all zeroes, which means we can't represent negative numbers.

B. Correct implementation:

```
typedef unsigned packed_t;

int xbyte(packed_t word, int bytenum) {
  // cut off unwanted bytes to the left
  int shifted = word << ((3 - bytenum) << 3);
  // cut off unwanted bytes to the right
  // number will be sign extended because type is int
  return shifted >> 24;
}
```

Problem 2.72

A. Because `sizeof` returns a `size_t` (which is unsigned), the result of `maxbytes - sizeof(val)` is also unsigned and thus always ≥ 0.

B. Correct implementation:

```
void copy_int(int val, void *buf, int maxbytes) {
  if (maxbytes >= sizeof(val))
    memcpy(buf, (void *)&val, sizeof(val));
}
```

Problem 2.73

```
int saturating_add(int x, int y) {
  int sum = x + y;
  int offset = (sizeof(int) << 3) - 1;

  // create masks with all 0s or all 1s
  int x_neg = x >> offset;
  int y_neg = y >> offset;
  int sum_neg = sum >> offset;

  int pos_overflow = ~x_neg & ~y_neg & sum_neg;
  int neg_overflow = x_neg & y_neg & ~sum_neg;

  // use masks to choose between sum and bounds
  return (INT32_MAX & pos_overflow) | (INT32_MIN & neg_overflow) |
         (sum & ~(pos_overflow | neg_overflow));
}
```

Problem 2.74

```
int tsub_ok(int x, int y) {
  int offset = (sizeof(int) << 3) - 1;
  int x_neg = x >> offset;
  int y_neg = y >> offset;
  int diff_neg = (x - y) >> offset;

  int pos_overflow = ~x_neg & y_neg & diff_neg;
  int neg_overflow = x_neg & ~y_neg & ~diff_neg;
  return !(pos_overflow | neg_overflow);
}
```

Problem 2.75

```
unsigned unsigned_high_prod(unsigned x, unsigned y) {
  unsigned prod = signed_high_prod(x, y);
  unsigned offset = (sizeof(unsigned) << 3) - 1;
  unsigned x_sign = x >> offset;
  unsigned y_sign = y >> offset;
  return prod + x_sign * y + y_sign * x;
}
```

Problem 2.76

```
void *calloc(size_t nmemb, size_t size) {
  if (nmemb == 0 || size == 0) {
    return NULL;
  }

  size_t total_size = nmemb * size;

  if (total_size / size != nmemb) {
    // Overflow occurred
    return NULL;
  }

  void *ptr = malloc(total_size);

  if (ptr == NULL) {
    // Allocation failed
    return NULL;
  }

  memset(ptr, 0, total_size);
  return ptr;
}
```

Problem 2.77

```
int times_17 = (x << 4) + x;
int times_neg7 = x - (x << 3);
int times_60 = (x << 6) - (x << 2);
int times_neg112 = (x << 4) - (x << 7);
```

Problem 2.78

```
int divide_power(int x, int k) {
  int w = sizeof(int) << 3;
  int mask = x >> (w - 1);
  int bias = mask & ((1 << k) - 1);
  return (x + bias) >> k;
}
```

Problem 2.79

```
int mul3div4(int x) {
  int mul3 = (x << 1) + x;
  int w = sizeof(int) << 3;
  int mask = mul3 >> (w - 1);
  int bias = mask & 3;
  return (mul3 + bias) >> 2;
}
```

Problem 2.80

```
int threefourths(int x) {
  // calculate result for higher bits that are not relevant for rounding
  int high_div4 = x >> 2;
  int high_result = (high_div4 << 1) + high_div4;

  // for the remainder we can safely multiply first since it cannot overflow
  int rem = x - (high_div4 << 2);
  int rem_mul3 = (rem << 1) + rem;
  int mask = x >> ((sizeof(int) << 3) - 1);
  int bias = mask & 3;
  int low_result = (rem_mul3 + bias) >> 2;

  return high_result + low_result;
}
```

Problem 2.81

```c
int main() {
  // example values
  int k = 9;
  int j = 5;

  int A = -1 << k;
  int B = ((1 << k) - 1) << j;
}
```

Problem 2.82

A. `false` for $x = -2^{31}$ which is TMin_{32} because $-\text{TMin} = \text{TMin}$ (since it overflows).

B. Always `true`. If any overflows happen they will be the same on both sides. (ring properties of two's complement arithmetic)

C. Always `true`. `~x` is equal to `-x - 1`. So we can write it as `-x - 1 - y - 1 + 1 = -(x+y) - 1` which simplifies to `-x - y - 1 = -x - y - 1`.

D. Always `true`. By negating we turn $y - x$ into $x - y$. Regardless of the negation, both sides are treated as unsigned for the comparison. And unsigned and two's complement arithmetic are the same on the bit level.

E. Always `true`. This basically sets the 2 lower bits to 0. These bits have a positive weight regardless of the sign bit, so setting them to 0 can not make the number larger.

Problem 2.83

A. Let V be the value of the string. Shifting the binary point k to the right results in y.yyyy... which equals both $V + Y$ and $V \times 2^k$. These can now be equated:

$$\begin{aligned} V + Y &= V \times 2^k \\ Y &= V \times 2^k - V \\ Y &= V\left(2^k - 1\right) \\ V &= \frac{Y}{2^k - 1} \end{aligned}$$

B.

(a) $V = \frac{5}{2^3 - 1} = \frac{5}{7}$

(b) $V = \frac{6}{2^4 - 1} = \frac{2}{5}$

(c) $V = \frac{19}{2^6 - 1} = \frac{19}{63}$

Problem 2.84

```
return ((ux << 1) == 0 && (uy << 1) == 0) // +0 and -0 are considered equal
       || (!sx && !sy && ux <= uy)        // both positive
       || (sx && sy && ux >= uy)          // both negative
       || (sx > sy);                      // different signs
```

Problem 2.85

A. The number 7.0 will have exponent $E = 2$, significand $M = 1.11_2 = \frac{7}{4}$, fraction $f = 0.11_2 = \frac{3}{4}$ and value $V = 7$. The exponent will be represented as `10...01` (bias is $2^{k-1} - 1 = 01...1_2$) and the fraction as `110...0`.

B. The largest odd integer that can be represented exactly will have exponent $E = n$, significand $M = 1.1...1 = 2 - \frac{1}{2^n}$, fraction $f = 0.1...1_2 = 1 - \frac{1}{2^n}$ and value $V = 2^{n+1} - 1$. The exponent will be the binary representation of $n + 2^{k-1} - 1$ and the fraction will be represented as $1...1$.

C. The smallest positive normalized value has exponent $2 - 2^{k-1}$ and fraction 0, giving a value of $1.0 \times 2^{2-2^{k-1}}$. The reciprocal of this is $V = 2^{2^{k-1}-2}$. It will have exponent $E = 2^{k-1} - 2$, significand $M = 1$ and fraction $f = 0$. The biased exponent will be $2^{k-1} - 2 - 2^{k-1} - 1 = -3$ represented by `1...101` and the fraction by `0...0`.

Problem 2.86

	Extended precision	
Description	Value	Decimal
Smallest positive denormalized	$2^{-63} \times 2^{2-2^{14}}$	2.645×10^{-4951}
Smallest positive normalized	$2^{2-2^{14}}$	3.362×10^{-4932}
Largest normalized	$(2 - 2^{-63}) \times 2^{2^{14}-1}$	1.190×10^{4932}

Problem 2.87

Description	Hex	M	E	V	D
-0	`0x8000`	0	-14	-0	-0.0
Smallest value > 2	`0x4001`	$\frac{1025}{1024}$	1	1025×2^{-9}	2.001953125
512	`0x6000`	1	9	512	512.0
Largest denormalized	`0x03FF`	$\frac{1023}{1024}$	-14	1023×2^{-24}	0.0000609756
$-\infty$	`0xFC00`	-	-	$-\infty$	$-\infty$

Description	Hex	M	E	V	D
Number with hex representation `3BB0`	`0x3BB0`	$\frac{123}{64}$	-1	123×2^{-7}	0.9609375

Problem 2.88

Format A		Format B	
Bits	Value	Bits	Value
`1 01111 001`	$\frac{-9}{8}$	`1 0111 0010`	$\frac{-9}{8}$
`0 10110 011`	176	`0 1110 0110`	176
`1 00111 010`	$\frac{-5}{1024}$	`1 0000 0101`	$\frac{-5}{1024}$
`0 00000 111`	$\frac{7}{2^{17}}$	`0 0000 0001`	$\frac{1}{1024}$
`1 11100 000`	-2^{13}	`1 1110 1111`	-248
`0 10111 100`	384	`0 1111 0000`	∞

Problem 2.89

A. Always `true`. `float` cannot represent every 32-bit integer value in full precision, but `double` can, so they will be rounded in the same way.

B. `false` for $x = 0$ and $y = \text{TMin}_{32}$. The integer result will overflow before it is cast to `double` while the operation with `double`s has a wider range and does not overflow.

C. Always `true`. The result of adding two floats in the 32-bit integer range cannot overflow.

D. (Apparently not always true, cannot reproduce solution tho)

E. `false` for $x = 1$ and $z = 0$. Dividing by 0 results in `NaN`.

Problem 2.90

```
float fpwr2(int x) {
  /* Result exponent and fraction */
  unsigned exp, frac;
  unsigned u;

  if (exp < -126-23) {
    /* Too small. Return 0.0 */
    exp = 0;
    frac = 0;
  } else if (exp < -126) {
    /* Demornalized result */
    exp = 0;
    frac = 1 << (exp + 126 + 23);
  } else if (exp <= 127) {
    /* Normalized result */
    exp = exp + 127;
    frac = 0;
  } else {
    /* Too big. Return +inf */
    exp = 0xFF;
    frac = 0;
  }

  /* Pack exp and frac into 32 bits */
  u = exp << 23 | frac;
  /* Return as float */
  return u2f(u);
}
```

Problem 2.91

A. Binary representation: `0 10000000 10010010000111111011011`. So the exponent is $128 - 127 = 1$ and the fractional number is $11.00100100001111110110 11_2$.

B. $\frac{22}{7} = 3\frac{1}{7} = 11.001001001001001\ldots_2$

C. The two approximations diverge starting from the 9th bit after the binary point.

Problem 2.92

```
typedef unsigned float_bits;

float_bits float_negate(float_bits f) {
  unsigned exp = f >> 23 & 0xFF;
  unsigned frac = f & 0x7FFFFF;

  if (exp == 0xFF && frac != 0) {
    // NaN, return as is
    return f;
  };

  unsigned mask = 1 << 31;
  return f ^ mask;
}
```

Problem 2.93

```
typedef unsigned float_bits;

float_bits float_absval(float_bits f) {
  unsigned exp = f >> 23 & 0xFF;
  unsigned frac = f & 0x7FFFFF;

  if (exp == 0xFF && frac != 0) {
    // NaN, return as is
    return f;
  };

  unsigned mask = (1U << 31) - 1;
  return f & mask;
}
```

Problem 2.94

```
typedef unsigned float_bits;

float_bits float_twice(float_bits f) {
  unsigned sign = f >> 31;
  unsigned exp = (f >> 23) & 0xFF;
  unsigned frac = f & 0x7FFFFF;

  if (exp == 0xFF) {
    // NaN, +-Infinity, return as is
    return f;
  };

  if (exp == 0) {
    // Denormalized number
    frac <<= 1;

    if (frac > 0x7FFFFF) {
      // normalize
      exp = 1;
      frac &= 0x7FFFFF;
    }
    return sign << 31 | (exp << 23) | frac;
  }

  // normalized number
  exp += 1;

  if (exp == 0xFF) {
    // overflow to infinity
    frac = 0;
  }
  return sign << 31 | (exp << 23) | frac;
}
```

Problem 2.95

```
typedef unsigned float_bits;

float_bits float_half(float_bits f) {
  unsigned sign = f >> 31;
  unsigned exp = (f >> 23) & 0xFF;
  unsigned frac = f & 0x7FFFFF;

  if (exp == 0xFF) {
    // NaN, +-Infinity, return as is
    return f;
  };

  // round to even
  unsigned roundup = (frac & 0x3) == 3;

  if (exp == 0) {
    // Denormalized number
    frac >>= 1;
    frac += roundup;
    return sign << 31 | (exp << 23) | frac;
  }

  // normalized number
  exp -= 1;

  if (exp == 0) {
    // denormalize
    frac |= 0x800000; // add implicit leading 1
    frac >>= 1;
    frac += roundup;
  }

  return sign << 31 | (exp << 23) | frac;
}
```

Problem 2.96

```
typedef unsigned float_bits;

int float_f2i(float_bits f) {
  unsigned sign = f >> 31;
  unsigned exp = (f >> 23) & 0xFF;
  unsigned frac = f & 0x7FFFFF;

  if (exp == 0xFF) {
    // NaN or infinity
    return 0x80000000;
  }

  if (exp == 0) {
    // Denormalized number
    return 0;
  }

  // Normalized number
  int unbiased_exp = exp - 127;

  if (unbiased_exp < 0) {
    // Absolute value less than 1
    return 0;
  }

  if (unbiased_exp >= 31) {
    // Overflow
    return 0x80000000;
  }

  if (unbiased_exp < 23) {
    frac >>= 23 - unbiased_exp;
  } else {
    frac <<= unbiased_exp - 23;
  }

  return (sign ? -1 : 1) * ((1 << unbiased_exp) + frac);
}
```

Problem 2.97

```c
typedef unsigned float_bits;

float_bits float_i2f(int i) {
  unsigned sign = i & (1U << 31);
  unsigned abs = i < 0 ? -i : i;

  if (abs == 0) {
    return 0;
  }

  // normalized number
  unsigned exp = 127 + 31;

  // shift left until bit 31 is set (will become implicit leading 1)
  while ((abs & (1U << 31)) == 0) {
    abs <<= 1;
    exp--;
  }

  unsigned frac = (abs >> 8) & 0x7FFFFF;
  unsigned residue = abs & 0xFF;

  // round to even
  if (residue > 0x80 || residue == 0x80 && (frac & 1)) {
    frac++;

    if (frac > 0x7FFFFF) {
      // renormalize
      frac &= 0x7FFFFF;
      frac >>= 1;
      exp++;
    }
  }

  return sign | (exp << 23) | frac;
}
```

Chapter 3 Machine-Level Representation of Programs

Problem 3.1

Operand	Value
`%rax`	`0x100`
`0x104`	`0xAB`
`$0x108`	`0x108`
`(%rax)`	`0xFF`

Operand	Value
`4(%rax)`	`0xAB`
`9(%rax,%rdx)`	`0x11`
`260(%rcx,%rdx)`	`0x13`
`0xFC(,%rcx,4)`	`0xFF`
`(%rax,%rdx,4)`	`0x11`

Problem 3.2

`movl`

`movw`

`movb`

`movb`

`movq`

`movw`

Problem 3.3

Instruction	Mistake
`movb $0xF, (%ebx)`	`%ebx` is a 32-bit register, memory addresses have to be 64 bit
`movl %rax, (%rsp)`	`%rax` is 64 bit and thus must be used with `movq`
`movw (%rax),4(%rsp)`	both operands are memory locations
`movb %al,%sl`	`%sl` is not a valid register
`movq %rax,$0x123`	destination cannot be an immediate value
`movl %eax,%rdx`	`%rdx` is 64 bit and thus must be used with `movq`
`movb %si, 8(%rbp)`	`%si` is 16 bit and thus must be used with `movw`

Problem 3.4

`src_t`	`dest_t`	**Instruction**
long	long	`movq (%rdi), %rax` `movq %rax, (%rsi)`
char	int	`movsbl (%rdi), %eax` `movl %eax, (%rsi)`
char	unsigned	`movsbl (%rdi), %eax` `movl %eax, (%rsi)`
unsigned char	long	`movzbl (%rdi) %eax` `movq %rax (%rsi)`
int	char	`movl (%rdi) %eax` `movb %al (%rsi)`
unsigned	unsigned char	`movl (%rdi) %eax` `movb %al (%rsi)`
char	short	`movsbw (%rdi) %ax` `movw %ax (%rsi)`

Problem 3.5

```
void decode1(long *xp, long *yp, long *zp) {
  long x = *xp;
  long y = *yp;
  long z = *zp;
  *yp = x;
  *zp = y;
  *xp = z;
}
```

Problem 3.6

Instruction	**Result**
`leaq 6(%rax), %rdx`	$x + 6$
`leaq (%rax,%rcx), %rdx`	$x + y$

Instruction	Result
`leaq (%rax,%rcx,4), %rdx`	$x + 4y$
`leaq 7(%rax,%rax,8), %rdx`	$9x + 7$
`leaq 0xA(,%rcx,4), %rdx`	$4y + 10$
`leaq 9(%rax,%rcx,2), %rdx`	$x + 2y + 9$

Problem 3.7

```
long t = 5 * x + 2 * y + 8 * z;
```

Problem 3.8

Instruction	Destination	Value
`addq %rcx,(%rax)`	`0x100`	`0x100`
`subq %rdx,8(%rax)`	`0x108`	`0xA8`
`imulq $16,(%rax,%rdx,8)`	`0x118`	`0x110`
`incq 16(%rax)`	`0x110`	`0x14`
`decq %rcx`	`%rcx`	`0x0`
`subq %rdx,%rax`	`%rax`	`0xFD`

Problem 3.9

C	Assembly
`x <<= 4`	`salq $4, %rax`
`x >>= n`	`sarq %cl, %rax`

Problem 3.10

```
long arith2(long x, long y, long z) {
  long t1 = x | y;
  long t2 = t1 >> 3;
  long t3 = ~t2;
  long t4 = z - t3;
  return t4;
}
```

Problem 3.11

A. It sets the `rdx` register to all zeroes.

B. `movq $0, %rdx` is a more straightforward way to express this.

C. The `xor` instruction only takes 3 bytes to encode, while `mov` takes 7 bytes. (Check using `echo 'movq $0, %rdx' | as -o /tmp/test.o | objdump -d /tmp/test.o`)

Problem 3.12

```
void uremdiv(unsigned long x, unsigned long y,
             unsigned long *qp, unsigned long *rp)
x in %rdi, y in %rsi, qp in %rdx, rp in %rcx

uremdiv:
  movq %rdx, %r8      Copy qp
  movq %rdi, %rax     Move x to lower 8 bytes of dividend
  xorq %rdx, %rdx     Set upper 8 bytes of dividend to 0
  divq %rsi           Divide by y (unsigned)
  movq %rax, (%r8)    Store quotient at qp
  movq %rdx, (%rcx)   Store remainder at rp
  ret
```

Problem 3.13

Part	`data_t`	`COMP`
A	`int`	`<`
B	`short`	`>=`
C	`unsigned char`	`<=`
D	`(unsigned) long` / some pointer	`!=`

Problem 3.14

Part	`data_t`	`TEST`
A	`long`	`>=`
B	`(unsigned) short`	`==`
C	`unsigned char`	`>`
D	`int`	`<=`

Problem 3.15

A. `4003fe`

B. `400425`

C. Address of `ja` is `400543`, address of `pop` is `400545`.

D. `400560`

Problem 3.16

A.

```
void cond(long a, long *p) {
  if (p == 0)
    goto done;
  if (*p >= a)
    goto done;
  *p = a;
done:
  return;
}
```

B. The assembly code contains two conditional branches because the condition in C consists of two parts. If the first part fails, the second part will be skipped (since they are connected by `&&`).

Problem 3.17

A.

```
long gotodiff_se(long x, long y) {
  long result;
  if (x < y)
    goto x_lt_y;
  ge_cnt++;
  result = x - y;
  return result;
x_lt_y:
  lt_cnt++;
  result = y - x;
  return result;
}
```

B. They are mostly equivalent. However, the first form is more convenient if there is no else case.

Problem 3.18

```
long test(long x, long y, long z) {
  long val = x + y + z;
  if (x < -3) {
    if (y < z)
      val = x * y;
    else
      val = y * z;
  } else if (x > 2)
    val = x * z;
  return val;
}
```

Problem 3.19

A.

$$T_{\mathrm{MP}} = 2(T_{\mathrm{ran}} - T_{\mathrm{OK}})$$
$$T_{\mathrm{MP}} = 2(31 - 16) = 30$$

B.

$$T_{\mathrm{OK}} + T_{\mathrm{MP}} = 16 + 30 = 46$$

Problem 3.20

A. `OP` is division

B.

```
arith:
  leaq    7(%rdi), %rax     store x + 7 in return register
  testq   %rdi, %rdi        set condition codes based on x
  cmovns  %rdi, %rax        set return value to x if x >= 0
  sarq    $3, %rax          divide by 8 (arithmetic shift)
  ret
```

For negative numbers we add a bias to round towards 0 instead of -inf. When dividing by 2^k the bias is $2^k - 1$.

Problem 3.21

```
long test(long x, long y) {
  long val = 8*x;
  if (y > 0) {
    if (x < y)
      val = y-x;
    else
      val = x&y;
  } else if (y <= -2)
    val = x+y;
  return val;
}
```

Problem 3.22

A. The maximum value of n for which we can represent $n!$ with a 32-bit int is 12.

B. The maximum value of n for which we can represent $n!$ with a 64-bit long is 20.

This can be verified with the following code:

```
int main() {
  long result = 1;
  long prev_result = 1;
  int factor = 1;

  for (;;) {
    result *= ++factor;
    printf("%d) %ld\n", factor, result);

    if (result / factor != prev_result) {
      printf("Overflow detected at factor %d\n", factor);
      break;
    }
    prev_result = result;
  }
}
```

Problem 3.23

A.

C Variable	Register
`x`	`%rdi` (initially) and `%rax`
`y`	`%rcx`
`n`	`%rdx`

B. The pointer always points to `x` so `(*p)++` just increments `x`. This is then combined with `x += y` into a single `leaq` instruction.

C.

```
dw_loop:
  movq    %rdi, %rax            put x into the return register
  movq    %rdi, %rcx
  imulq   %rdi, %rcx            y = x*x
  leaq    (%rdi,%rdi), %rdx     n = 2*x
.L2
  leaq    1(%rcx,%rax), %rax    x += y + 1
  subq    $1, %rdx              n--
  testq   %rdx, %rdx            test n
  jg    .L2                     loop if n > 0
  rep; ret                      return x
```

Problem 3.24

```
long loop_while(long a, long b) {
  long result = 1;
  while (a < b) {
    result = result * (a+b);
    a = a + 1;
  }
  return result;
}
```

Problem 3.25

```
long loop_while2(long a, long b) {
  long result = b;
  while (b > 0) {
    result = result * a;
    b = b - a;
  }
  return result;
}
```

Problem 3.26

A. jump-to-middle translation was used for the loop

B.

```
long fun_a(unsigned long x) {
  long val = 0;
  while (x) {
    val ^= x;
    x >>= 1;
  }
  return val & 0x1;
}
```

C. The function computes the parity of the number of on bits in `x`, returning 1 if it's an odd number, returning 0 otherwise.

Problem 3.27

```
long fact_for_while(long n) {
  long result = 1;
  if (n < 2)
    goto done;
  long i = 2;
loop:
  result *= i;
  i++;
  if (i <= n)
    goto loop;
done:
  return result;
}
```

Problem 3.28

A.

```
long fun_b(unsigned long x) {
  long val = 0;
  long i;
  for (i = 64; i; i--) {
    val *= 2;
    val |= (x & 0x1);
    x >>= 1;
  }
  return val;
}
```

B. The compiler knows that the loop will always run exactly 64 times, so the initial test and jump can be omitted.

C. This funciton reverses the bit representation of `x`.

Problem 3.29

A. We would also skip incrementing i, resulting in an infinite loop.

```
long sum = 0;
long i = 0;

while (i < 10) {
  if (i & 1)
    continue; /* causes infinite loop */
  sum += i;
  i++;
}
```

B. We can fix it like this:

```
long sum = 0;
long i = 0;

while (i < 10) {
  if (i & 1)
    goto end_loop;
  sum += i;
end_loop:
  i++;
}
```

Problem 3.30

A. Case labels: -1, 0, 1, 2, 4, 5, 7

B. 0 and 7 as well as 2 and 4 label the same cases.

Problem 3.31

```
void switcher(long a, long b, long c, long *dest) {
  long val;
  switch(a) {
  case 5:
    c = b ^ 15;
    /* Fall through */
  case 0:
    val = c + 112;
    break;
  case 2:
  case 7:
    val = (b + c) << 2;
    break;
  case 4:
    val = a;
    break;
  default:
    val = b;
  }
  *dest = val;
}
```

Problem 3.32

Label	PC	Instruction	%rdi	%rsi	%rax	%rsp	*%rsp	Description
M1	0x400560	callq	10	—	—	0x7fffffffe820	-	Call first(10)
F1	0x400548	lea	10	—	—	0x7fffffffe818	0x400565	Entry of first
F2	0x40054c	sub	10	11	—	0x7fffffffe818	0x400565	Prepare argument for last
F3	0x400550	callq	9	11	—	0x7fffffffe818	0x400565	Call last(9, 11)
L1	0x400540	mov	9	11	—	0x7fffffffe810	0x400555	Entry of last
L2	0x400543	imul	9	11	9	0x7fffffffe810	0x400555	Calculate return value
L3	0x400547	retq	9	11	99	0x7fffffffe810	0x400555	Return 99 from last
F4	0x400555	repz retq	9	11	99	0x7fffffffe818	0x400565	Return 99 from first
M2	0x400565	mov	9	11	99	0x7fffffffe820	-	Resume main

Problem 3.33

The params are (in order):

Param	Type	Register
a	`int`	`%edi`
b	`short`	`%si`
u	`long *`	`%rdx`
v	`char *`	`%rcx`

It's also possible to swap `a` and `u` with `b` and `v`, respectively.

Problem 3.34

A. Local values `a0` - `a5` are stored in callee-saved registers.

B. Local values `a6` and `a7` are stored on the stack.

C. Not all local values are stored in callee-saved registers because there are only six of them (`%r12` - `%r15`, `%rbp` and `rbx`).

Problem 3.35

A. The value of the parameter `x` is stored in callee-saved register `%rbx`.

B.

```
long rfun(unsigned long x) {
  if (x == 0)
    return 0;
  unsigned long nx = x >> 2;
  long rv = rfun(nx);
  return x + rv;
}
```

Problem 3.36

Array	Element size	Total size	Start address	Element i
S	2	14	x_S	$x_S + 2i$
T	8	24	x_T	$x_T + 8i$
U	8	48	x_U	$x_U + 8i$
V	4	32	x_V	$x_V + 4i$
W	8	32	x_W	$x_W + 8i$

Problem 3.37

Expression	Type	Value	Assembly code
`S+1`	`short *`	$x_S + 2$	`leaq 2(%rdx), %rax`
`S[3]`	`short`	$M[x_S + 6]$	`movw 6(%rdx), %ax`
`&S[i]`	`short *`	$x_S + 2i$	`leaq (%rdx,%rcx,2), %rax`
`S[4*i+1]`	`short`	$M[x_S + 8i + 2]$	`movw 2(%rdx,%rcx,8), %ax`
`S+i-5`	`short *`	$x_S + 2i - 10$	`leaq -10(%rdx,%rcx,2), %rax`

Problem 3.38

```
long sum_element(long i, long j)
i in %rdi, j in %rsi
sum_element:
  leaq    0(,%rdi,8), %rdx        8*i
  subq    %rdi, %rdx              7*i
  addq    %rsi, %rdx              7*i+j
  leaq    (%rsi,%rsi,4), %rax     5*j
  addq    %rax, %rdi              i+5*j
  moveq   Q(,%rdi,8), %rax        Q[i+5*j]
  addq    P(,%rdx,9), %rax        Q[i+5*j] + P[7*i+j]
```

Based on the assembly code the values are $M = 5$ and $N = 7$.

Problem 3.39

The address of element i, j in a nested array is

$$\&\mathrm{A}[i][j] = x_A + L(C \cdot i + j)$$

where L is the element size in bytes and C is the number of columns.

Applying this to our pointers:

- `Aptr:` $\&\mathrm{A}[i][0] = x_A + 4(16i + 0) = x_A + 64i$
- `Bptr:` $\&\mathrm{B}[0][k] = x_B + 4(16 \cdot 0 + k) = x_B + 4k$
- `Bend:` $\&\mathrm{B}[16][k] = x_B + 4(16 \cdot 16 + k) = x_B + 1024 + 4k$

Problem 3.40

```
void fix_set_diag_opt(fix_matrix A, int val) {
  int *Aptr = &A[0][0];
  long i = 0;
  long iend = N(N+1);
  do {
    Aptr[i] = val;
    i += (N+1); // advance by one row and one col
  } while (i != iend);
}
```

Problem 3.41

A.

Field	Offset
`p`	0
`s.x`	8
`s.y`	12
`next`	16

B. In total the structure requires 24 bytes.

C.

```
void sp_init(struct prob *sp) {
  sp->s.x  = sp->s.y;
  sp->p    = &(sp->s);
  sp->next = sp;
}
```

Problem 3.42

A.

```
long fun(struct ELE *ptr) {
  long sum = 0;
  while (ptr) {
    sum += ptr->v;
    ptr = ptr->p;
  }
  return sum;
}
```

B. `ELE` implements a linked list and `fun` computes the sum of all elements.

Problem 3.43

expr	*type*	Code
`up->t1.u`	`long`	`movq (%rdi), %rax` `movq %rax, (%rsi)`
`up->t1.v`	`short`	`movw 8(%rdi), %ax` `movw %ax, (%rsi)`
`&up->t1.w`	`char *`	`leaq 10(%rdi), %rax` `movq %rax, (%rsi)`
`up->t2.a`	`int *`	`movq %rdi, (%rsi)`

`up->t2.a[up->t1.u]`	`int`	`movq (%rdi), %rax`
		`movl (%rdi,%rax,4) %eax`
		`movl %eax, (%rsi)`

`*up->t2.p`	`char`	`movq 8(%rdi), %rax`
		`movb (%rax), %al`
		`movb %al, (%rsi)`

Problem 3.44

A.

i	c	j	d	**Size**	**Alignment**
0	4	8	12	16	4

B.

i	c	d	j	**Size**	**Alignment**
0	4	5	8	16	8

C.

w	c	**Size**	**Alignment**
0	6	10	2

D.

w	c	**Size**	**Alignment**
0	16	40	8

E.

a	t	**Size**	**Alignment**
0	24	40	8

Problem 3.45

A. & B.

a	b	c	d	e	f	g	h	**Size**
0	8	16	24	28	32	40	48	56

C. We can rearrange the fields like this to minimize wasted space:

a	b	d	f	e	c	g	h	**Size**
0	8	10	11	12	16	24	32	40

Problem 3.46

A.

Value	**Label**
`00 00 00 00 00 40 00 76`	Return address
`01 23 45 67 89 AB CD EF`	Value of callee-saved register %rbx
	buf = %rsp

B.

Value	**Label**
`00 00 00 00 00 40 00 34`	Return address
`33 32 31 30 39 38 37 36`	Value of callee-saved register %rbx
`35 34 33 32 31 30 39 38`	
`37 36 35 34 33 32 31 30`	buf = %rsp

C. The program attempts to return to the address `0x400034`. The two low bytes were overwritten by `0x34` (last char from stdin) and `0x00` (terminating null character).

D. The register %rbx will be corrupted after returning, since its saved value on the stack was overritten and it is callee-saved, so the callee is not allowed to change it.

E. `strlen` returns the length without the null terminator, so `malloc` allocates one too few bytes. Also, `malloc` can return `NULL`, which is not safe to pass into `strcpy`.

Problem 3.47

A. `0xd754` - `0xb754` = `0x2000` = $2 \cdot 16^3 = 2^{13}$.

B. It would take about $\frac{2^{13}}{2^7} = 2^6 = 64$ attempts.

Problem 3.48

A.

Without protector:

Name	Position in stack frame
`buf`	`(%rsp)`
`v`	`24(%rsp)`
canary value	-

With protector:

Name	Position in stack frame
`buf`	`16(%rsp)`
`v`	`8(%rsp)`
canary value	`40(%rsp)`

B. In the rearranged variant the canary value is immediately adjacent to `buf`, so `v` cannot be corrupted.

Problem 3.49

A. s_2 is calculated by subtracting $8n + 22$ rounded down to the closest multiple of 16 from s_1. If n is odd it will result in $s_1 - (8n + 8)$, otherwise $s_1 - (8n + 16)$.

B. We take $s_2 + 7$ and round down to the closest multiple of 8 (alignment). This effectively rounds *up* s_2 to the closest multiple of 8.

C.

n	s_1	s_2	p	e_1	e_2
5	2065	2017	2024	1	7
6	2064	2000	2000	16	0

D. s_2 is aligned to multiples of 16, and p is aligned to multiples of 8.

Problem 3.50

Expression	Mapping
`val1`	`d`
`val2`	`i`
`val3`	`l`
`val4`	`f`

Problem 3.51

T_x	T_y	Instruction(s)
long	double	`vcvtsi2sdq %rdi, %xmm0, %xmm0`
double	int	`vcvttsd2si %xmm0, %eax`
double	float	`vunpcklpd %xmm0, %xmm0, %xmm0` `vcvtpd2ps %xmm0, %xmm0`
long	float	`vcvtsi2ssq %rdi, %xmm0, %xmm0`
float	long	`vcvttss2siq %xmm0, %rax`

Problem 3.52

A.

```
double g1(double a, long b, float c, int d);
```

Argument	Register
a	%xmm0
b	%rdi
c	%xmm1
d	%esi

B.

```
double g2(int a, double *b, float *c, long d);
```

Argument	Register
a	%edi
b	%rsi

Argument	Register
c	%rdx
d	%rcx

C.

```
double g3(double *a, double b, int c, float d);
```

Argument	Register
a	%rdi
b	%xmm0
c	%esi
d	%xmm1

D.

```
double g4(float a, int *b, float c, double d);
```

Argument	Register
a	%xmm0
b	%rdi
c	%xmm1
d	%xmm2

Problem 3.53

arg1_t	arg2_t	arg3_t	arg4_t
int	float	long	double
int	long	float	double

Problem 3.54

```
double funct2(double w, int x, float y, long z) {
  return y*x - w/z;
}
```

Problem 3.55

Low-order 4 bytes:

```
00 00 00 00
```

High-order 4 bytes:

```
40 40 00 00
```

Complete number in binary:

```
01000000 01000000 00000000 00000000 00000000 00000000 00000000 00000000
seeeeeee eeeeffff ffffffff ffffffff ffffffff ffffffff ffffffff ffffffff
```

So the sign is positive.

The exponent is $1028 - 1023 = 5$.

The fraction is 1.0 (with implied leading 1).

The value is thus $2^5 \times 1.0 = 32.0$.

Problem 3.56

A.

Constant: `0x7fffffffffffffff`

Binary: `01111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111`

This bitmask applied with `&` calculates the absolute value of the `double`.

B.

This sets the register to all zeroes since `a ^ a = 0`.

C.

Constant: `0x8000000000000000`

This bitmask applied with `^` negates the value of the `double`.

Problem 3.57

```
double funct3(int *ap, double b, long c, float *dp) {
  int a = *ap;
  float d = *dp;
  if (a < b) {
    return c*d;
  } else {
    return c + 2*d;
  }
}
```

Problem 3.58

```
long decode2(long x, long y, long z) {
  y -= z;
  x *= y;
  return ((y << 63) >> 63) ^ x;
}
```

Problem 3.59

```
dest in %rdi, x in %rsi, y in %rdx
store_prod:
  movq    %rdx, %rax        y = yl
  cqto                      %rdx:%rax = yh:yl
  movq    %rsi, %rcx
  sarq    $63, %rcx         xh = sign of x
  imulq   %rax, %rcx        r = yl * xh
  imulq   %rsi, %rdx        s = yh * xl
  addq    %rdx, %rcx        r + s
  mulq    %rsi              %rdx:%rax = xl * yl = t
  addq    %rcx, %rdx        r + s + th
  movq    %rax, (%rdi)      store pl in memory
  movq    %rdx, 8(%rdi)     store ph in memory
  ret
```

`r` and `s` are corrections for signed multiplication which we have to consider because `mulq` does unsigned multiplication (and only the the lower 64 bits (= size of factors) are equivalent between two's complement and unsigned).

Problem 3.60

A.

Value	Register
`x`	`%rdi`
`n`	`%esi`
`result`	`%rax`
`mask`	`%rdx`

B. The initial values of `result` and `mask` are 0 and 1, respectively.

C. The test condition is `mask != 0`.

D. `mask` is arithmetically shifted left by `n`.

E. The result is OR'ed with `x & mask`.

F.

```
long loop(long x, int n) {
  long result = 0;
  long mask;
  for (mask = 1; mask != 0; mask <<= n) {
    result |= (x & mask);
  }
  return result;
}
```

Problem 3.61

```
long cread(long *xp) {
  long zero = 0;
  if (!xp) xp = &zero;
  return *xp;
}
```

Problem 3.62

```
typedef enum {MODE_A, MODE_B, MODE_C, MODE_D, MODE_E} mode_t;

long switch3(long *p1, long *p2, mode_t action) {
  long result = 0;
  switch(action) {
    case MODE_A:
      result = *p2;
      *p2 = *p1;
      break;
    case MODE_B:
      *p1 += *p2;
      result = *p1;
      break;
    case MODE_C:
      *p1 = 59;
      result = *p2;
      break;
    case MODE_D:
      *p1 = *p2;
      /* fall through */
    case MODE_E:
      result = 27;
      break;
    default:
      result = 12;
  }
  return result;
}
```

Problem 3.63

```
long switch_prob(long x, long n) {
  long result = x;
  switch(n) {
    case 60:
    case 62:
      result <<= 3;
      break;
    case 63:
      result >>= 3;
      break;
    case 64:
      result *= 15;
      /* fall through */
    case 65:
      result *= result;
      /* fall through */
    case 61:
      result += 75;
  }
  return result;
}
```

Problem 3.64

A. Formula for location of array element `A[i][j][k]`:

$$\&\mathrm{D}[\mathrm{i}][\mathrm{j}][\mathrm{k}] = x_D + 8(iST + jT + k)$$

where R, S and T are the sizes of the nested arrays: `long D[R][S][T]`.

B.

```
long store_ele(long i, long j, long k, long *dest)
i in %rdi, j in %rsi, k in %rdx, dest in %rcx
store_ele:
  leaq    (%rsi,%rsi,2), %rax     3*j
  leaq    (%rsi,%rax,4), %rax     4(3*j)+j = 13*j
  movq    %rdi, %rsi              j = i
  salq    $6, %rsi                64*i
  addq    %rsi, %rdi              65*i
  addq    %rax, %rdi              13*j + 65*i
  addq    %rdi, %rdx              13*j + 65*i + k
  movq    A(,%rdx,8), %rax        read array element
  movq    %rax, (%rcx)            store at dest
  movl    $3640, %eax             return sizeof(A)
  ret
```

The index is calculated as $5 \cdot 13 \cdot i + 13 \cdot j + k$ which suggests that $S = 5$ and $T = 13$.

We also know that $8RST = 3640$, so $R = \frac{3640}{8 \cdot 5 \cdot 13} = 7$.

Problem 3.65

A. The pointer to element `A[i][j]` is in `%rdx`.

B. The pointer to element `A[j][i]` is in `%rax`.

C. `%rax` is incremented by $120 = 15 \cdot 8$ every iteration, so each row must have 15 elements, thus $M = 15$.

Problem 3.66

The address that we read each iteration (`%rcx`) is incremented by $8(4n + 1)$ so `NC` must be defined as $4n + 1$.

The loop condition compares `i` (in register `%rdx`) with `NR(n)` (in register `%rdi`). Looking at `%rdi` we know that `NR(n)` must be $3n$.

Problem 3.67

A.

Offset from %rsp	Description
96	
88	
80	`r.q`
72	`r.u[1]`
64	`r.u[0]`
56	
48	
40	
32	
24	`z`
16	`s.p`
8	`s.a[1]`
0	`s.a[0]`

B. `eval` passes a pointer to the space on the stack allocated for `r` to `process`.

C. Elements of `s` are accessed by calculating the address offsets for each element. It starts at `%rsp+8` because the return address is stored at `%rsp`.

D. Values on the return structure are set based on the address passed in as `%rdi`.

E. (see **A.**)

F. Structure function agruments are passed as pointers on the stack. Space for a function return value that is a structure is allocated by the caller and a pointer to it is passed as a hidden argument.

Problem 3.68

We see that to access `t` an offset of 8 is used. `t` is of type `int` so it must be aligned to an offset divisible by 4. This means B must be between 5 and 8.

Similarly, to access `u` an offset of 32 is used. `s` starts at offset 12. So `s` and possible padding take up 20 bytes. `u` is of type `long` which means it must be aligned to an offset divisible by 8. `s` has type `short` so each element takes up 2 bytes. From this we conclude that A must be between 7 and 10.

From the offset used to access `y` we see that `x` plus possible padding takes up 184 bytes. `y` has to be aligned to an offset divisible by 8. `x` itself is of type `int` so each element uses 4 bytes. This means the array itself must be either 184 bytes long or 180 bytes plus 4 bytes of padding. Which means it has either 45 or 46 elements. The only possible way to get either of these by multiplying possible values for A and B is $9 \cdot 5 = 45$.

So we can conclude that $A = 9$ and $B = 5$.

Problem 3.69

```
i in %rdi, bp in %rsi
test:
  mov     0x120(%rsi),%ecx        read last (offset 288)
  add     (%rsi),%ecx             first + last
  lea     (%rdi,%rdi,4),%rax      5*i
  lea     (%rsi,%rax,8),%rax      8*5*i
  mov     0x8(%rax),%rdx          read idx
  movslq  %ecx,%rcx               extend to 64 bits
  mov     %rcx,0x10(%rax,%rdx,8)
  retq
```

A. The size of `a_struct` is 40 bytes. Using the offset of `last` we have $\text{CNT} = \frac{288-8}{40} = 7$.

B.

```
typedef struct {
  long idx;
  long x[4];
} a_struct;
```

Problem 3.70

A.

Field	Offset
`e1.p`	0
`e1.y`	8
`e2.x`	0
`e2.next`	8

B. In total the structure requires 16 bytes.

C.

```
up in %rdi
proc:
  movq    8(%rdi), %rax     read e2.next (or e1.y)
  movq    (%rax), %rdx      *e2.next
  movq    (%rdx), %rdx      *(e2.next)->p
  subq    8(%rax), %rdx     subtract e2.next->e1.y
  movq    %rdx, (%rdi)      store in e2.x
  ret
```

```
union ele {
  struct {
    long *p;
    long y;
  } e1;
  struct {
    long x;
    union ele *next;
  } e2;
};

void proc (union ele *up) {
  up->e2.x = *(up->e2.next->e1.p) - up->e2.next->e1.y;
}
```

Problem 3.71

```
#include <stdio.h>
#include <string.h>

#define BUFSIZE 64

void good_echo() {
  char buf[BUFSIZE];
  while (fgets(buf, BUFSIZE, stdin)) {
    if (fputs(buf, stdout) == EOF)
      return; // error while writing

    int i;
    for (i = 0; buf[i] && buf[i] != '\n'; i++)
      ;
    if (buf[i] == '\n')
      return;

    // or using standard library
    // if (strchr(buf, '\n'))
    //   return;
  }
}
```

Problem 3.72

```
long vframe(long n, long idx, long *q) {
  long i;
  long *p[n];
  p[0] = &i;
  for (i = 1; i < n; i++) {
    p[i] = q;
  }
  return *p[idx];
}
```

A. We calculate $8n + 30$ and then round down to the closest multiple of 16. For even n we get $8n + 16$ and for odd n we get $8n + 24$. s_2 is calculated by subtracting this amount from s_1.

B. p is computed by rounding s_2 up to the closest multiple of 16 (by adding 15 and then rounding down).

C. *skipped*

D. *skipped*

Problem 3.73

```
.globl find_range
find_range:
  vxorps %xmm1, %xmm1, %xmm1  # set %xmm1 = 0
  vucomiss %xmm1, %xmm0
  jp .OTHER
  jb .NEG
  ja .POS
  movq $1, %rax
  ret
.NEG:
  movq $0, %rax
  ret
.POS:
  movq $2, %rax
  ret
.OTHER:
  movq $3, %rax
  ret
```

Test all possible inputs by calling from C:

```
#include <stdint.h>
#include <stdio.h>

typedef enum { NEG, ZERO, POS, OTHER } range_t;

extern range_t find_range(float x);

long r[4];

int main() {
  uint32_t bits = 0;
  while (1) {
    float f = *((float *)&bits); // preserve bits instead of value
    r[find_range(f)]++;
    if (bits == UINT32_MAX)
      break;
    bits++;
  }

  printf("NEG  : %ld\n", r[0]);
  printf("ZERO : %ld\n", r[1]);
  printf("POS  : %ld\n", r[2]);
  printf("OTHER: %ld\n", r[3]);
  printf("TOTAL: %ld\n", r[0] + r[1] + r[2] + r[3]);
}
```

Problem 3.74

```
.globl find_range
find_range:
  vxorps %xmm1, %xmm1, %xmm1  # set %xmm1 = 0
  vucomiss %xmm1, %xmm0
  movq $1, %rax
  movq $0, %rdi
  cmovb %rdi, %rax
  movq $2, %rdi
  cmova %rdi, %rax
  movq $3, %rdi
  cmovp %rdi, %rax
  ret
```

(use C code above for testing)

Problem 3.75

A. Complex arguments are passed in two registers, one containing the real and one containing the imaginary part.

B. Similarly for return values: real part in `%xmm0`, imaginary part in `%xmm1`.

Chapter 4 Processor Architecture

Problem 4.1

Instruction	Encoding
`irmovq $15,%rbx`	`30 f3 0f 00 00 00 00 00 00 00`
`rrmovq %rbx,%rcx`	`20 31`
`rmmovq %rcx,-3(%rbx)`	`40 13 fd ff ff ff ff ff ff ff`
`addq  %rbx,%rcx`	`60 31`
`jmp loop`	`70 0c 01 00 00 00 00 00 00`

Problem 4.2

Byte Sequence	Instruction Sequence
A.	
`0x100: 30f3fcffffffffffffff`	`  irmovq $-4,%rbx`
`0x10a: 4063000800000000000`	`  rmmovq %rsi,2048(%rbx)`

Byte Sequence	Instruction Sequence
`0x114: 00`	`  halt`
B.	
`0x200: a06f`	`  pushq %rsi`
`0x202: 800c02000000000000`	`  call proc`
`0x20b: 00`	`  halt`
`0x20c:`	`proc:`
`0x20c: 30f30a00000000000000`	`  irmovq $10,%rbx`
`0x216: 90`	`  ret`
C.	
`0x300: 5054070000000000000`	`  mrmovq 7(%rsp),%rbp`
`0x30a: 10`	`  nop`
`0x30b: f0`	invalid instruction
`0x30c: b01f`	`  popq %rcx`
D.	
`0x400:`	`loop:`
`0x400: 6113`	`  subq %rcx,%rbx`
`0x402: 730004000000000000`	`  je loop`
`0x40c: 00`	`  halt`
E.	
`0x500: 6362`	`  xorq %rsi,%rdx`
`0x502: a0f0`	`f` means "no register" but `pushq` requires one (but the second part must be `f`)

Problem 4.3

```
long sum(long *start, long count)
start in %rdi, count in %rsi
sum:
  xorq %rax,%rax        sum = 0
  andq %rsi,%rsi        Set CC
  jmp test              Goto test
loop:
  mrmovq (%rdi),%r10    Get *start
  addq %r10,%rax        Add to sum
  iaddq $8,%rdi         start++
  iaddq $-1,%rsi        count-- and set CC
test:
  jne loop              Stop when 0
  ret
```

Problem 4.4

```
long rsum(long *start, long count)
start in %rdi, count in %rsi
rsum:
  irmovq $0,%rax        sum = 0
  andq %rsi,%rsi        Set CC
  jg rec                Stop if count <= 0
  ret
rec:
  pushq %rbx            Save callee-saved register
  mrmovq (%rdi),%rbx    Read *start
  irmovq $1,%r8
  subq %r8,%rsi         count--
  irmovq $8,%r8
  addq %r8,%rdi         start++
  call rsum             Recurse
  addq %rbx,%rax        sum += *start
  popq %rbx             Restore callee-saved register
  ret
```

Problem 4.5

```
long absSum(long *start, long count)
start in %rdi, count in %rsi
absSum:
  irmovq $8,%r8         Constant 8
  irmovq $1,%r9         Constant 1
  xorq %rax,%rax        sum = 0
  andq %rsi,%rsi        Set CC
  jmp test              Goto test
loop:
  mrmovq (%rdi),%r10    Get *start
  andq %r10,%r10        Set CC
  jl sub
add:
  addq %r10,%rax        Add positive number to sum
  jmp cont
sub:
  subq %r10,%rax        Subtract negative number from sum
cont:
  addq %r8,%rdi         start++
  subq %r9,%rsi         count-- and set CC
test:
  jne loop              Stop when 0
  ret
```

Problem 4.6

```
long absSum(long *start, long count)
start in %rdi, count in %rsi
absSum:
  irmovq $8,%r8         Constant 8
  irmovq $1,%r9         Constant 1
  xorq %rax,%rax        sum = 0
  andq %rsi,%rsi        Set CC
  jmp test              Goto test
loop:
  mrmovq (%rdi),%r10    x = *start
  xorq %r11,%r11        Set 0
  subq %r10,%r11        -x
  cmovg %r11,%r10       if -x > 0 then x = -x
  addq %r10,%rax        Add to sum
  addq %r8,%rdi         start++
  subq %r9,%rsi         count-- and set CC
test:
  jne loop              Stop when 0
  ret
```

Problem 4.7

It implies that the register value is read before decrementing the stack pointer.

Problem 4.8

It implies that the value read from memory is written to `%rsp` instead of incrementing it. It is equivalent to:

```
mrmovq (%rsp),%rsp
```

Problem 4.9

We can define `xor` like this:

```
bool xor = (a && !b) || (!a && b)
```

It relates to `eq` like this:

```
bool xor = !eq
```

Problem 4.10

We can implement word-level equality using `xor` very similarly to the `==` implementation. We combine 64 `xor` circuits comparing each individual bit of `a` and `b`, combine all the outputs with `or`, and then negate the result.

Problem 4.11

```
word Min3 = [
  A <= B && A <= C : A;
  A <= C           : B;
  1                : C;
]
```

Using `B <= C` as the second condition also works.

Problem 4.12

Circuit for computing the median of three numbers:

```
word Med3 = [
  B <= A && A <= C : A;
  C <= A && A <= B : A;
  A <= B && B <= C : B;
  C <= B && B <= A : B;
  1                : C;
]
```

Problem 4.13

Stage	Generic	Specific
	`irmovq V, rB`	`irmovq $128, %rsp`
Fetch	icode:ifun $\leftarrow M_1[\text{PC}]$ rA:rB $\leftarrow M_1[\text{PC}+1]$ valC $\leftarrow M_8[\text{PC}+2]$ valP $\leftarrow \text{PC}+10$	icode:ifun $\leftarrow M_1[\text{0x016}] = 3:0$ rA:rB $\leftarrow M_1[\text{0x017}] = \text{f}:4$ valC $\leftarrow M_8[\text{0x018}] = 128$ valP $\leftarrow \text{0x016}+10 = \text{0x020}$
Decode	-	-
Execute	valE $\leftarrow 0 + \text{valC}$	valE $\leftarrow 0 + 128 = 128$
Memory	-	-
Write back	R[rB] $\leftarrow$ valE	R[%rsp] $\leftarrow 128$
PC update	PC $\leftarrow$ valP	PC $\leftarrow$ 0x020

The instruction sets `%rsp` to 128 and increments the program counter by 10.

Problem 4.14

Stage	Generic	Specific
	`popq rA`	`popq %rax`
Fetch	icode:ifun $\leftarrow M_1[\text{PC}]$ rA:rB $\leftarrow M_1[\text{PC}+1]$ valP $\leftarrow \text{PC}+2$	icode:ifun $\leftarrow M_1[\text{0x02c}] = \text{b}:0$ rA:rB $\leftarrow M_1[\text{0x02d}] = 0:\text{f}$ valP $\leftarrow$ 0x02e
Decode	valA $\leftarrow$ R[%rsp] valB $\leftarrow$ R[%rsp]	valA $\leftarrow$ R[%rsp] $= 120$ valB $\leftarrow$ R[%rsp] $= 120$
Execute	valE $\leftarrow \text{valB}+8$	valE $\leftarrow 120+8 = 128$
Memory	valM $\leftarrow M_8[\text{valA}]$	valM $\leftarrow M_8[120] = 9$
Write back	R[%rsp] $\leftarrow$ valE R[rA] $\leftarrow$ valM	R[%rsp] $\leftarrow 128$ R[%rax] $\leftarrow 9$
PC update	PC $\leftarrow$ valP	PC $\leftarrow$ 0x02e

The instruction sets `%rax` to 9, `%rsp` to 128 and increments the program counter by 2.

Problem 4.15

Effect of `pushq %rsp`:

- push the old stack pointer to the stack (using the new (decremented) pointer as address)
- decrement stack pointer by 8
- increment the PC by 2

This conforms to the desired behavior of Y86-64, since it pushes the old value to the stack.

Problem 4.16

`popq %rsp` has the effect of storing the value from the stack in `%rsp` (not the decremented stack pointer). This is the desired behavior for Y86-64.

Problem 4.17

Stage	`cmovXX rA, rB`
Fetch	icode:ifun $\leftarrow M_1[\text{PC}]$ rA:rB $\leftarrow M_1[\text{PC}+1]$ valP $\leftarrow \text{PC}+2$
Decode	valA $\leftarrow$ R[rA]
Execute	valE $\leftarrow 0+\text{valA}$ Cnd $\leftarrow$ Cond(CC, ifun)
Memory	-
Write back	if(Cnd) R[rB] $\leftarrow$ valE
PC update	PC $\leftarrow$ valP

Problem 4.18

Stage	**Generic**	**Specific**
	`call Dest`	`call 0x041`
Fetch	icode:ifun $\leftarrow M_1[\text{PC}]$ valC $\leftarrow M_8[\text{PC}+1]$ valP $\leftarrow \text{PC}+9$	icode:ifun $\leftarrow M_1[0\text{x}037] = 8{:}0$ valC $\leftarrow M_8[0\text{x}038] = 0\text{x}041$ valP $\leftarrow 0\text{x}040$
Decode	valB $\leftarrow$ R[%rsp]	valB $\leftarrow$ R[%rsp] $= 128$
Execute	valE $\leftarrow \text{valB}+(-8)$	valE $\leftarrow 128+(-8) = 120$
Memory	$M_8[\text{valE}] \leftarrow \text{valP}$	$M_8[120] \leftarrow 0\text{x}040$
Write back	R[%rsp] $\leftarrow$ valE	R[%rsp] $\leftarrow 120$
PC update	PC $\leftarrow$ valC	PC $\leftarrow 0\text{x}041$

This instruction decreases the stack pointer by 8, sets the PC to the destination address of the procedure to call and pushes the return address to the stack.

Problem 4.19

```
bool need_calC = icode in { IIRMOVQ, IRMMOVQ, IMRMOVQ, IJXX, ICALL };
```

Problem 4.20

```
word srcB = [
  icode in { IRMMOVQ, IMRMOVQ, IOPQ } : rB;
  icode in { IPUSHQ, IPOPQ, ICALL, IRET } : RRSP;
  1 : RNONE;
]
```

Problem 4.21

```
word dstM = [
  icode in { IMRMOVQ, IPOPQ } : rA;
  1 : RNONE;
]
```

Problem 4.22

Port M should have priority over port E. This way the instruction `popq %rsp` will have the effect of writing the value read from memory to `%rsp`.

Problem 4.23

```
word aluB = [
  icode in { IOPQ, IRMMOVQ, IMRMOVQ, IPUSHQ, IPOPQ, ICALL, IRET } : valB;
  icode in { IRRMOVQ, IIRMOVQ } : 0;
];
```

Problem 4.24

```
word dstE = [
  icode in { IRRMOVQ } && Cnd : rB;
  icode in { IIRMOVQ, IOPQ } : rB;
  icode in { IPUSHQ, IPOPQ, ICALL, IRET } : RRSP;
  1 : RNONE;
];
```

Problem 4.25

```
word mem_data = [
  icode in { IRMMOVQ, IPUSHQ } : valA;
  icode in { ICALL } : valP;
];
```

Problem 4.26

```
bool mem_write = icode in { IRMMOVQ, IPUSHQ, ICALL };
```

Problem 4.27

```
word Stat = [
  imem_error || dmem_error : SADR;
  !instr_valid : SINS;
  icode == IHALT : SHLT;
  1 : SAOK;
];
```

Problem 4.28

A. The blocks should be distributed as evenly as possible to the two stages. If we insert a register between C and D, stage 1 will take 170 ps and stage 2 will take 130 ps. So the total cycle time is $170 + 20 = 190$ ps. This gives a throughput of:

$$\frac{1 \text{ instruction}}{190 \text{ ps}} \times \frac{1000 \text{ ps}}{1 \text{ ns}} \approx 5.26 \text{ GIPS}$$

Since it takes 2 cycles for an instruction to complete the latency is $190 \text{ ps} \times 2 = 380$ ps.

B. The blocks can be divided into 3 stages like this: `[A, B] [C, D] [E, F]`. This gives a cycle time of 130 ps. So the throughput is:

$$\frac{1 \text{ instruction}}{130 \text{ ps}} \times \frac{1000 \text{ ps}}{1 \text{ ns}} \approx 7.69 \text{ GIPS}$$

with a latency of $130 \text{ ps} \times 3 = 390 \text{ ps}$.

C. Dividing into 4 stages like this: `[A] [B, C] [D] [E, F]` gives a cylce time of 110 ps. So the throughput is 9.09 GIPS with latency 440 ps.

D. With 5 stages we can achieve the minimum cycle time (which leads to maximum throughput). Dividing stages like `[A] [B] [C] [D] [E, F]` gives a cycle time of 100 ps. So the throughput is 10 GIPS with latency 500 ps.

Problem 4.29

A.

Latency (ps): $20k + 300$

Throughput (GIPS): $\frac{1000}{\frac{300}{k}+20}$

B. The limit of the throughput is $\lim_{k\to\infty} \frac{1000}{\frac{300}{k}+20} = 50$

Problem 4.30

```
word f_stat = [
  imem_error : SADR;
  !instr_valid : SINS;
  f_icode == IHALT : SHLT;
  1 : SAOK;
];
```

Problem 4.31

```
word d_dstE = [
  D_icode in { IRRMOVQ } : D_rB;
  D_icode in { IIRMOVQ, IOPQ } : D_rB;
  D_icode in { IPUSHQ, IPOPQ, ICALL, IRET } : RRSP;
  1 : RNONE;
];
```

Problem 4.32

It writes the address stored in `%rsp` (incremented stack pointer) to `%rax`.

Problem 4.33

We can use a similar test program as in the previous problem, only add two `nop` instructions to make sure the `popq` instruction is in the write-back stage while the `rrmovq` instruction is in the decode stage. Again, the incremented stack pointer will be written instead of the value read from memory.

Problem 4.34

```
word d_valB = [
  d_srcB == e_dstE : e_valE;
  d_srcB == M_dstM : m_valM;
  d_srcB == M_dstE : M_valE;
  d_srcB == W_dstM : W_valM;
  d_srcB == W_dstE : W_valE;
  1 : d_rvalB;
];
```

Problem 4.35

Using `E_dstE` instead of `e_dstE` would not consider the condition codes, so a conditional move with a failed contition would be executed anyway.

```
irmovq $2, %rax
irmovq $3, %rdx
xorq %rcx, %rcx
cmovqne %rdx, %rax    # failed condition, move should not be executed
addq %rax, %rax       # should be 4, not 6
```

Problem 4.36

```
word m_stat = [
  dmem_error : SADR;
  1 : M_stat;
];
```

Problem 4.37

```
  call proc
  # should not reach this
proc:
  xorq %rcx, %rcx
  jne target
  irmovq $2, %rax
  halt
target:
  ret
```

Problem 4.38

```
  irmovq mem, %rbx
  mrmovq 0(%rbx), %rsp
  ret
  halt
rtnpt:
  irmovq $5, %rsi # ret should return here
  halt

.pos 0x40
mem:
  .quad stack # holds stack pointer

.pos 0x50
stack:
  .quad rtnpt # holds return point
```